Sanjay M

ic23btech11018@iith.ac.in | zachardyyy0110@gmail.com | +91 8610235650 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.63
The Velammal Int School, Panchetti, Tiruvallur, Tamil Nadu	Class XII (Central Board of Secondary Education)	2022	89.00%
Velammal Matric Hr Sec School, Panchetti, Tiruvallur, Tamil Nadu	Class X (State Board of Secondary Educations, Tamil Nadu)	2020	90.80%

SCHOLASTIC ACHIEVEMENTS

- **NTSE Scholar (2020):** Cleared Level 2 of the prestigious National Talent Search Examination (Top 0.01% nationwide), earning a recurring merit-based academic scholarship.
- **Winner, D-CoDE Hackathon (Oct, 2025):** Awarded 1st Place (out of 50+ graduate/Ph.D. teams) for designing an engineering-first chemical product model and secured incubation funding for further scaling.

PROJECTS

High-Throughput Asynchronous Rate Limiter & API Gateway | *Python, FastAPI, Redis, Lua, Docker* | GitHub

- Architected an asynchronous L7 API Gateway middleware in Python (FastAPI) capable of enforcing rate-limiting policies across distributed microservices with **p99 latency < 2ms**.
- Implemented the **Token Bucket** rate-limiting algorithm using atomic **Lua scripts** executed directly inside Redis, eliminating race conditions during high-concurrency traffic bursts.
- Enforced standardized HTTP rate-limiting headers (X-RateLimit-Limit, X-RateLimit-Remaining, Retry-After) and automatic 429 Too Many Requests request dropping to protect backend database health.
- Containerized the gateway and Redis infrastructure using **Docker Compose** and benchmarked performance using concurrent traffic scripts to validate system stability under load.

Embedded LSM-Tree Storage Engine | *Python, File I/O, Systems Architecture, Pytest* | GitHub

- Engineered a two-tier **key-value storage engine** inspired by RocksDB, utilizing in-memory MemTables and disk-backed SSTables to convert random I/O into sequential appends.
- Optimized point-lookup latency to **O(log N)** by designing an in-memory sparse index mapping key byte offsets within sorted on-disk binary SSTable files.
- Implemented non-destructive tombstone deletions and K-Way merge compaction, consolidating multi-generational SSTables, resolving duplicates, and reclaiming disk space.

PolyMatch: PXRD Peak Analysis & Polymorph Engine | *Python, FastAPI, SciPy, React, Plotly.js* | GitHub

- Engineered a full-stack web application to automate powder X-ray diffraction spectrum comparison, replacing hours of manual laboratory analysis against patented drug forms.
- Developed an O(log N) binary-search peak-matching algorithm with SciPy signal processing to extract 2 local maxima and flag novel crystal polymorph anomalies (+ or -).
- Built an asynchronous FastAPI REST backend and React interactive dashboard, cutting diffractogram comparison and novel phase identification time to under 2 seconds.

SKILLS

Problem Solving: 500+ Data Structures & Algorithms problems solved across LeetCode, Codeforces, and GeeksforGeeks

Computer Science Fundamentals: Data Structures & Algorithms, Operating Systems, Database Management Systems (DBMS), Computer Networks

Languages: Python, C, C++, JavaScript

Backend & Systems: FastAPI, Node.js, Express, Redis, Docker, gRPC, Protobuf, REST APIs

Frontend & UI: React, Plotly.js, Axios

ML & Data Science: PyTorch, TensorFlow, Keras, Scikit-learn, SciPy, Pandas, NumPy, Matplotlib, Seaborn

Databases & Storage: MongoDB, PostgreSQL/SQL, LSM-Trees (Embedded Storage Engines)

Tools & Platforms: Git, VS Code, Jupyter Notebook, Google Colab, LaTeX, JWT

RELEVANT COURSES

Computer Science & Systems: Introduction to Programming (Language - C), Data Structures & Applications

Computational Science & ML: Intro to AI, Machine Learning in Chemistry, Numerical and Computational Chemistry, AI in Drug Discovery

Mathematics & Engineering: Linear Algebra, Calculus, Differential Equations, Probability & Statistics

POSITIONS OF RESPONSIBILITY

Torque | SciTech Club - Automobile and Design Club | Core Member

Cepheid | SciTech Club - Astronomy and Astrophysics Club | Outreach and Events Core

EML | Extra Mural Lectures Club | Logistics and Hospitality Coordinator

CERTIFICATIONS

Machine Learning Specialization | *DeepLearning.AI & Stanford University (Coursera)*

- Completed 3-course series by Andrew Ng covering Supervised Learning (Regression/Classification), Advanced Learning Algorithms (Neural Networks/Decision Trees), and Unsupervised Learning (Clustering, Recommenders, Reinforcement Learning).

EXTRACURRICULAR

- Volunteered in National Service Scheme(**NSS**) events